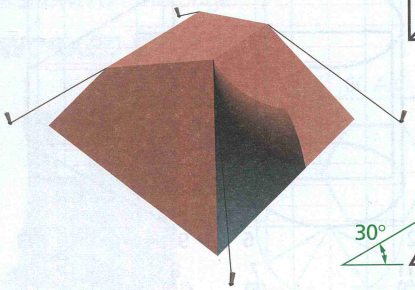
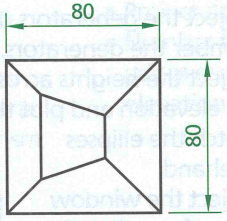
ORTHOGRAPHIC PROJECTION – PYRAMIDS AND TRUE SHAPES

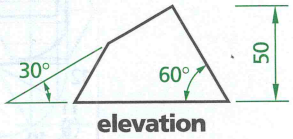
WORKED EXAMPLE 1

A child's play tent is to be made in the form of a cut-off square pyramid. Draw:

- the elevation and plan
- the end elevation on the left
- the surface development of the four sides
- the true shape of the top of the tent.

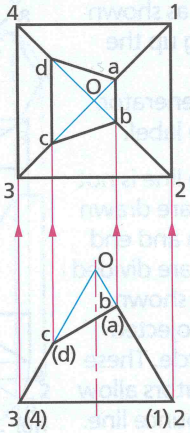






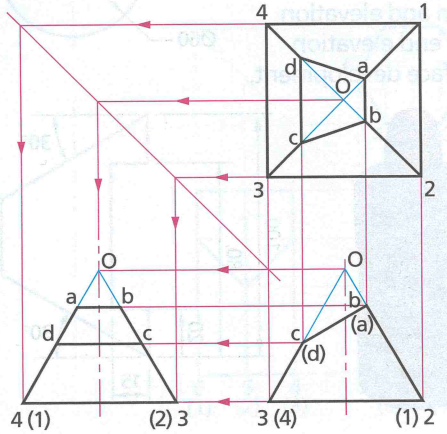
1. Draw the elevation and plan.

- Draw the elevation, including the apex 'O'.
- Project upwards and draw the plan.
- Number the four corners and the apex.
- Project points a, b, c and d up to the plan.



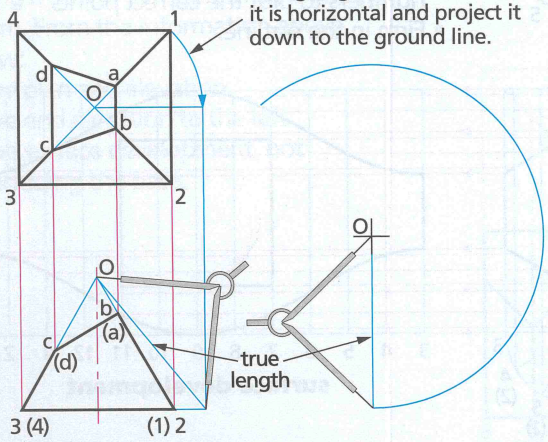
2. Draw the end elevation.

- Project heights across from the elevation.
- Use the bounce line to project breadths from the plan.
- Follow the numbers to plot the shape of the end elevation.



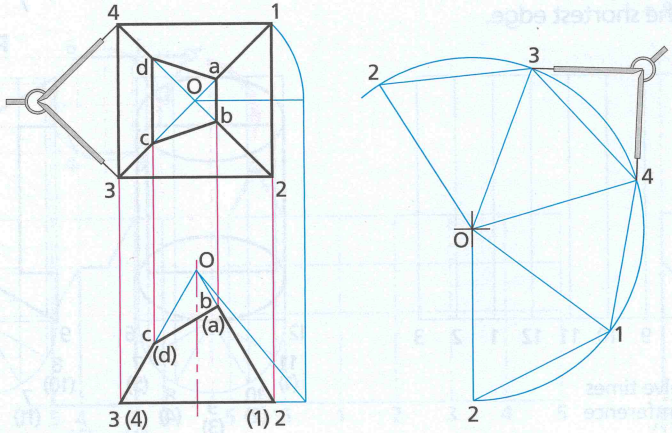
3. Find the true length of a complete edge.

- On the plan, swing edge O–1 around until it is horizontal.
- Drop this point down to the ground line.
- On the elevation, join this new point to the apex O. This **true length** becomes the radius for the surface development.



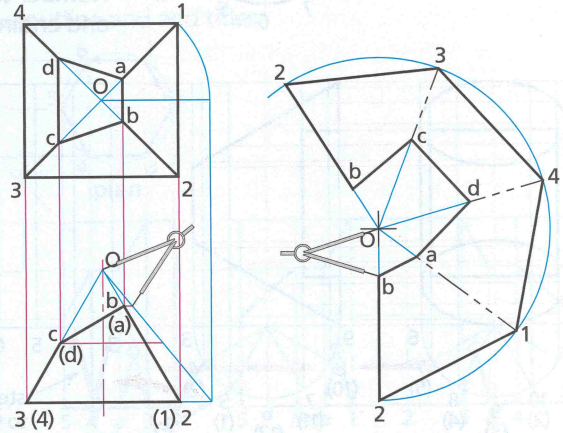
4. Step off the four sides.

- Step the base length around the arc four times.
- Number these points and join them to the centre (point O).



5. Complete the surface development.

- On the elevation, project points a, b, c and d across to the true length.
- Step these sizes onto the surface development.



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